

AZURE DATA FACTORY (ADF) – ULTIMATE INTERVIEW CHEAT SHEET (DATA ENGINEER)

(SHORT, HIGH-IMPACT, AND READY FOR LAST-MINUTE REVISION)

1. ADF CORE COMPONENTS

Pipeline → Logical grouping of activities

Activity → Single step (Copy, Lookup, ForEach...)

Dataset → Pointer to data structures (tables, files)

Linked Service → Connection string / credential

Integration Runtime (IR) → Compute for data movement & transformation

Triggers → Schedule/event-based pipeline execution

2. INTEGRATION RUNTIME TYPES

IR Type	Use Case	Example Scenario
Azure IR	Cloud copy + Data Flow	Copy ADLS → Synapse
Self-hosted IR	On-prem to cloud	On-prem SQL → Azure
Azure SSIS IR	Run SSIS packages	Lift-and-shift ETL

3. MOST ASKED IN INTERVIEWS – ADF ACTIVITIES

- **Copy Activity** → Moves data between stores
- **Lookup Activity** → Fetch metadata, select configs
- **ForEach Activity** → Loop through arrays/files
- **If Condition** → Branching logic
- **Get Metadata** → Detect file existence, size, modified date
- **Execute Pipeline** → Reusability
- **Data Flow** → Transformations at scale (Spark-based)

4. TRIGGERS

- **Schedule Trigger** → Every X time
- **Event Trigger** → File arrives in ADLS/Blob
- **Tumbling Window Trigger** → Late arrival, checkpoints, watermarking

Scenario:

“PROCESS EVERY 15-MIN BATCH AND HANDLE BACKFILL AUTOMATICALLY.” → Tumbling Window.

5. PARAMETERIZATION (CRITICAL FOR INTERVIEWS)

ADF allows **Pipeline** → **Dataset** → **Linked Service** parameter flow.

Common Use Case:

Build **one pipeline** to process **dynamic files**:

Example parameters:

- FileName
- SourcePath
- TargetTable

6. MAPPING DATA FLOWS (ADF DATA FLOW)

- **Derived Column**
- **Conditional Split**
- **Aggregate**
- **Join**
- **Surrogate Key**
- **Window**
- **Rank**

Important Note: Data Flow is **Spark-based**, runs on **Azure IR**.

7. ADF VS SYNAPSE PIPELINES

Feature	ADF	Synapse Pipelines
Purpose	ETL/ELT	Lakehouse + SQL + Spark
Data Flows	Yes	Yes
Orchestration	Strong	Strong
Integration	Azure-wide	Best with Synapse Workspace

ADF = ETL orchestrator

Synapse = ETL + Data Warehouse + Lakehouse

8. COPY ACTIVITY IN-DEPTH

- Supports **binary, tabular, parquet, JSON**
- Auto schema mapping
- **Staging** option for large loads (e.g., using Azure Blob)
- Supports **parallel copy, compression, column mapping**.

Scenario:

HOW TO SPEED UP 1 TB FILE COPY TO ADLS?

Answer: **Parallel copy, staging, partitioning, block blob optimization.**

9. ADF MONITORING

Key metrics:

- Pipeline Runs
- Activity Runs
- Trigger Runs
- Integration Runtime metrics

Failures → can be routed to Log Analytics / Alerts.

10. COMMON INTERVIEW SCENARIOS

✓ SCENARIO 1: DYNAMIC FILE INGESTION FROM ADLS EVERY DAY

Use:

- Event trigger
- Get Metadata → List files
- ForEach → Copy Activity
- Parameterized dataset

✓ SCENARIO 2: INCREMENTAL LOAD USING LAST MODIFIED DATE

Use:

- Lookup Activity → Fetch watermark from DB
- Filter files using Data Flow or pipeline expressions
- Update watermark after load

✓ SCENARIO 3: PROCESS ONLY NEW PARTITION FOLDERS

Use:

- Get Metadata (child items)
- Filter by **YYYY/MM/DD** folder

- ForEach → Copy/Transform

SCENARIO 4: CALL REST API WITH PAGINATION

Use:

- Web Activity
- Until Activity
- Append variable
- Copy Activity to sink in ADLS

11. EXPRESSIONS & FUNCTIONS (MOST USED)

- @utcnow()
- @concat()
- @formatDateTime()
- @activity('GetMetadata1').output.childItems
- @greater() / @less()
- @equals()

12. BEST PRACTICES (MAXIMUM INTERVIEWER WEIGHTAGE)

- Use **parameterized pipelines** (avoid hardcoding).
- Use **Key Vault** for secrets.
- Create **modular pipelines** (ingest, transform, load).
- Enable **retry policies** for unstable sources.
- Store metadata in a **config table** → drive dynamic pipelines.
- Enable **logging & exception handling** using **global logging pipelines**.

13. ADF PRICING (QUICK)

- **Orchestration cost** per activity/pipeline
- **Data Flow execution** (most expensive)
- **IR compute cost**
- **Pipeline trigger cost**